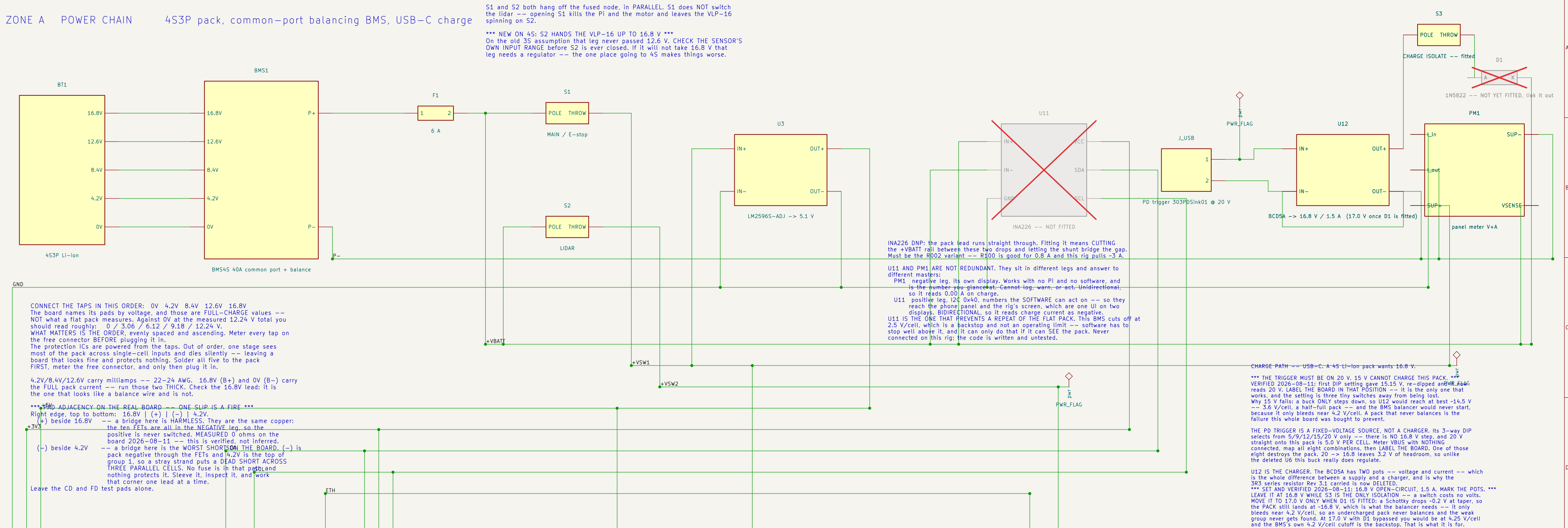
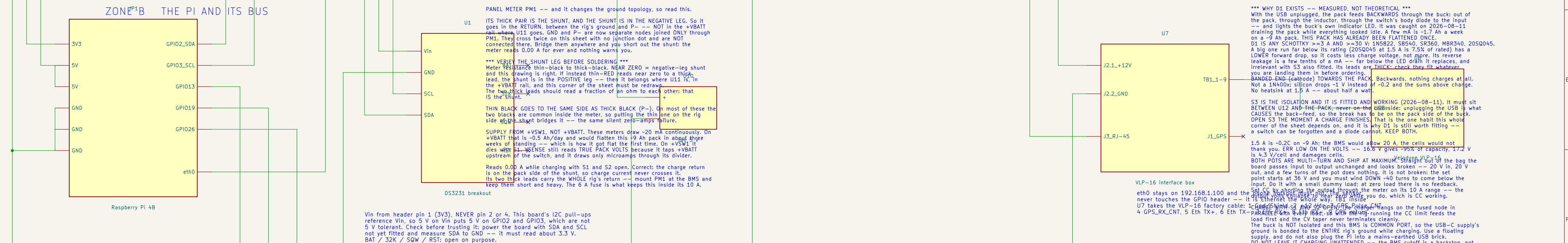


ZONE A POWER CHAIN 4S3P pack, common-port balancing BMS, USB-C charge



ZONE B THE PI AND ITS BUS



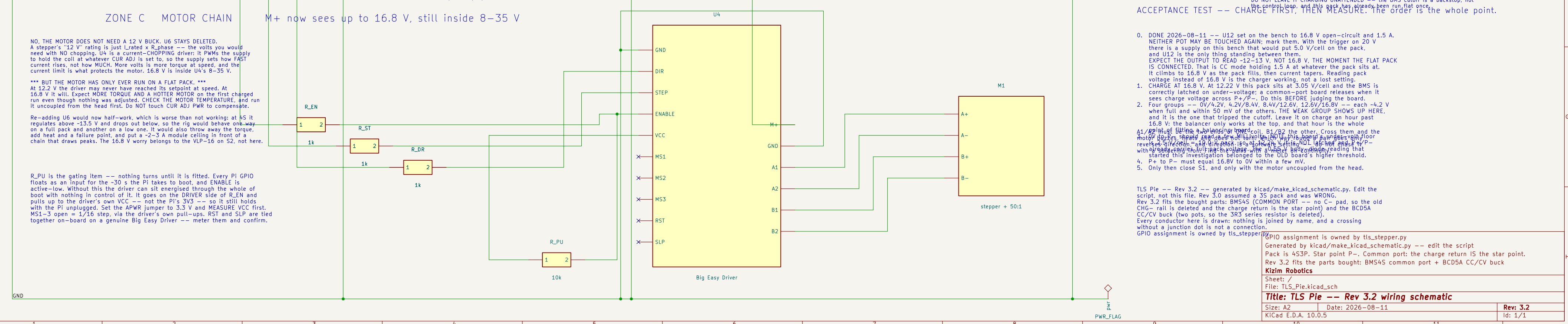
ZONE C MOTOR CHAIN

NO, THE MOTOR DOES NOT NEED A 12 V BUCK. U6 STAYS DELETED.
A stepper's "12 V" rating is just Rated $\times$ R-phase -- the volts you would need with NO chopping. U4 is a current-CHOPPING driver: it PWMs the supply to hold the coil at whatever CUR ADJ is set to, so the supply sets how FAST current rises, not how MUCH. More volts is more torque at speed, and the current limit is what protects the motor. 16.8 V is inside U4's 8-35 V.

*** BUT THE MOTOR HAS ONLY EVER RUN ON A FLAT PACK. ***
At 12.2 V the driver may never have reached its setpoint at speed. At 16.8 V it will. Expect MORE TORQUE AND A HOTTER MOTOR on the first charged run even though nothing was adjusted. CHECK THE MOTOR TEMPERATURE, and run it uncoupled from the head first. Do NOT touch CUR ADJ PWR to compensate.

Re-adding U6 would now half-work, which is worse than not working: at 45 it regulates above -13.5 V and drops out below, so the rig would behave one way on a full pack and another on a low one. It would also throw away the torque, add heat and a failure point, and put a -2-3 A module ceiling in front of a chain that draws peaks. The 16.8 V worry belongs to the VLP-16 on S2, not here

R-PU is the input item -- nothing turns it until it is fitted. Every PI, GPIO floats as the input for the ~30 s the PI takes to boot, and ENABLE is active-low. Without this the driver can sit energised through the whole of boot with nothing in control of it. It goes on the DRIVER side of R_EN and pulls up to the driver's own VCC -- not the PI's 3V3 -- so it still holds with the PI unplugged. Set the APWR jumper to 3.3 V and MEASURE VCC first. MS1-3 open = 1/16 step, via the driver's own pull-ups. RST and SLP are tied together on-board on a genuine Big Easy Driver -- meter them and confirm.



ACCEPTANCE TEST -- CHARGE FIRST, THEN MEASURE. The order is the whole point.

0. DONE 2026-08-11 — U12 set on the bench to 16.8 V open-circuit and 1.5 A. NEITHER POT MAY BE TOUCHED AGAIN; mark them. With the trigger on 20 V there is a supply on this bench that would put 5.0 V/cell on the pack, and U12 is the only thing standing between that and the battery. EXPECT THE OUTPUT TO READ -12~13 V. NOT 16.8 V. THE MOMENT THE FLAT PACK IS CONNECTED, THAT IS CC mode holding 1.5 A at whatever the pack sits at. It climbs to 16.8 V as the pack fills, then current tapers, until the pack voltage instead of 16.8 V is the charger working, not a lost setting.
1. CHARGE AT 16.8 V. At 12.22 v this pack sits at 3.05 V/cell and the BMS is correctly latched under-voltage; a common-pot board releases when it sees charge voltage across P+/P-, use this BEFORE JUDGING anything.
2. Four groups — 0V/4.2V, 4.2V/8.4V, 8.4V/12.6V, 12.6V/16.8V — each +4.2 V when full and within 0.2 mV of the others. THE WEAK GROUP SHOWS US HERE, and it is the one that tripped the cutoff. Leave it on charged, and after 16.8 V: the balancer only works at the top, and that hour is the whole day.
- A1/B1 must bring the balance back to C1/C2, the other. Cross them and the motor fires. ~~Should check~~ does not turn on until 16.8 V or higher. P+ has been reversed. ~~It's not the motor, it's the pack. So don't do anything with P-~~ with ~~motor and the flat pack voltage~~. The alarm beyond now reading that started this investigation belonged to the OLD board's higher threshold.
4. P+ to P- must equal 16.8V to 0V within a few mv.
5. Then only close S1, and only with the motor uncoupled from the head.

TLS Pie -- Rev 3.2 -- generated by kicad/make_kicad_schematic.py. Edit the script, not this file. Rev 3.0 assumed a 35 pin pack and was WRONG.
Rev 3.2 fits the bought parts: BM545 (COMMON PORT -- no C-pad, so the old CHG-- rail is deleted and the charge return is the star point) and the BC05A CC/CV buck (two pots, so the 3R3 series resistor is deleted).
Every conductor here is drawn: nothing is joined by name, and a crossing without a junction dot is not a connection.
GPIO assignment is owned by tls_stepper.py. GPIO assignment is owned by tls_stepper.py.

GPIO assignment is owned by `tls_stepper.py`

```
GPIO assignment is owned by tls_stepper.py
Generated by kicad/make_kicad_schematic.py -- edit the script
Pack is 453P. Star point P-. Common port: the charge return IS the star
Rev 3.2 fits the parts bought: BMS45 common port + BCD5A CC/CV bu
```

	Kizim Robotics
	Sheet: /
	File: TLS_Pie.kicad_sch

Title: TLS Pie -- Rev 3.2 wiring schematic

Size: A2	Date: 2026-08-11
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